

RESEARCH SERIES · CURRICULUM DESIGN

12-Week AI Upskill RoadMap

Outcome-based curriculum with free web resources, pre-read & post-read references for every session

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FORMAT

12 Sessions · 6 Phases + 5 Portfolio Projects**ABSTRACT**

This document provides the complete week-by-week curriculum for the IntelliForge AI Upskill RoadMap Programme. Each of the 12 sessions includes core topics, a concrete shipped deliverable, curated free pre-read resources (complete before the session), and post-read resources (deepen after). Five sessions culminate in portfolio-grade agentic AI projects with quantified business impact, mapped to real hiring signals. The curriculum spans six phases — AI Foundations, RAG & Data, Agents & Automation, Full-Stack AI, Production & DevOps, Capstone & Career — building to a live deployed AI SaaS product and a professional portfolio by Week 12.

INTELLIFORGE PRODUCT ECOSYSTEM**Upskill Platform**

upskill.intelliforge.tech

**Deep Research**

deep-research.intelliforge.tech

**HRMS**

hrms.intelliforge.tech

**Learning OS**

learning.intelliforge.tech



00

Curriculum At a Glance

Every session in this programme is outcome-based. You leave each week with one shipped artefact — not notes. By Week 12 you will have 12 real deliverables: code, live APIs, agents, deployed apps, and a personal brand. Five of these are portfolio-grade agentic systems that demonstrate hiring-signal engineering, not chatbot tutorials.

WEEK	MODULE	PHASE	DELIVERABLE TYPE
W01	AI & LLM Foundations	AI Foundations	Code
W02	AI Engineering Toolkit	AI Foundations	Code + Eval
W03	Vector Databases & Embeddings	RAG & Data	Data Pipeline
W04	RAG Systems Architecture	RAG & Data	Portfolio: RAG + Generator
W05	AI Agent Design Patterns	Agents & Automation	Portfolio: Agentic System
W06	Task Queues & Async AI Workflows	Agents & Automation	Portfolio: Multi-Agent
W07	Next.js 14 + AI Integration	Full-Stack AI	Portfolio: Swarm + SaaS
W08	Prisma, Supabase & Payments	Full-Stack AI	Monetised App
W09	Production Deployment & Observability	Production & DevOps	Runbook + Monitoring
W10	Security, Cost & Responsible AI	Production & DevOps	Portfolio: LLM Gateway
W11	Capstone — Build & Ship	Capstone	Shipped Product
W12	Career Strategy & Personal GTM	Career & GTM	Personal Brand

5 PORTFOLIO PROJECTS EMBEDDED IN CURRICULUM

W04 Research-to-Deck Generator · W05 Compliance Sentinel Agent · W06 Autonomous Code Reviewer · W07 Recruiting Swarm · W10 Cost-Aware Agent Router

Each solves an expensive real-world problem with quantified ROI. Full architecture breakdowns in Phase 7 (Pages 16–20).

INTELLIFORGE PRODUCT ECOSYSTEM — SKILLS FEED INTO THESE LIVE PRODUCTS

upskill.intelliforge.tech — The platform delivering this programme (W01–W12 bootcamp host) · deep-research.intelliforge.tech — AI research synthesis tool built on RAG (W03, W04, P3) · hrms.intelliforge.tech — AI-native HR & recruiting platform (W07, W08, P4) · learning.intelliforge.tech — Structured learning OS powering cohort journeys (W01, W11, W12)

12 curated open-source AI projects mapped to each curriculum phase — with exact contribution tasks, IntelliForge product alignment, and a 12-week contribution roadmap. Your merged PR is your most credible portfolio artefact.

12

Outcome-based sessions across 6 phases

36

Curated free resource links (pre + post read)

5

Portfolio-grade agentic AI projects with ROI



JOIN THE AI ROADMAP COHORT — WHATSAPP GROUP

Scan the QR or visit chat.whatsapp.com/LwxMCJIEqLm4oLOG0fLqme to join the cohort WhatsApp group. Get session updates, resource drops, peer support, and direct access to Girish.

W01

PHASE 1 · AI FOUNDATIONS

AI & LLM Foundations

How large language models actually work

POWERS →

- upskill.intelliforge.tech
- learning.intelliforge.tech

TOPICS COVERED

→ Transformer architecture: attention heads, tokens, embeddings

→ LLM training pipeline: pre-training, RLHF, instruction tuning

→ Prompt engineering: zero-shot, few-shot, chain-of-thought

→ Token economics: context windows, cost & latency tradeoffs

→ Claude API setup: messages API, system prompts, parameters

SESSION DELIVERABLE

Code

Build a CLI chatbot using the Claude API with a custom system prompt. Implement 3 prompt patterns (zero-shot, few-shot, chain-of-thought). Push to a public GitHub repo with a README documenting each pattern and sample output. This forms your first artefact on upskill.intelliforge.tech and your learning.intelliforge.tech profile.

↑ PRE-READ RESOURCES

Complete these before the session

01

Attention Is All You Need — Vaswani et al.
<https://arxiv.org/abs/1706.03762>

02

The Illustrated Transformer — Jay Alammar
<https://jalammar.github.io/illustrated-transformer/>

03

Intro to LLMs — Andrej Karpathy (YouTube 1hr)
https://www.youtube.com/watch?v=zjKBMFhNj_g

↓ POST-READ RESOURCES

Deepen your understanding after

01

Anthropic Claude API — Getting Started
<https://docs.anthropic.com/en/docs/intro-to-claude>

02

Prompt Engineering Guide — DAIR.AI
<https://www.promptingguide.ai/>

03

OpenAI Tokenizer Tool (interactive)
<https://platform.openai.com/tokenizer>

W02

PHASE 1 · AI FOUNDATIONS

AI Engineering Toolkit

SDKs, structured outputs, evals and cost control

POWERS →

- upskill.intelliforge.tech
- learning.intelliforge.tech

TOPICS COVERED

→ LangChain vs LlamaIndex vs raw SDK — when to use what

→ Structured outputs: JSON mode, function/tool calling, Zod validation

→ LLM evaluation: BLEU, ROUGE, LLM-as-judge patterns

→ Cost monitoring: token tracking, budget guardrails, caching

→ TypeScript + Python SDK patterns for production codebases

SESSION DELIVERABLE

Code + Eval

Wrap your W01 chatbot with structured JSON output using Zod. Build a 10-case eval harness scoring responses via LLM-as-judge. Output a markdown eval report with a pass/fail table and commentary on failure modes.

↑ PRE-READ RESOURCES

Complete these before the session

01

Anthropic Tool Use Guide — Official Docs

<https://docs.anthropic.com/en/docs/build-with-claude/tool-use>

02

LangChain Quickstart — Python

https://python.langchain.com/docs/get_started/quickstart

03

LLM Evaluation Metrics — Confident AI

<https://www.confident-ai.com/blog/llm-evaluation-metrics>

↓ POST-READ RESOURCES

Deepen your understanding after

01

OpenAI Evals Framework (GitHub)

<https://github.com/openai/evals>

02

Braintrust — LLM Eval Platform (free tier)

<https://www.braintrustdata.com/>

03

Zod Schema Validation Docs

<https://zod.dev/>

W03

PHASE 2 · RAG & DATA

Vector Databases & Embeddings

Semantic search infrastructure from first principles

POWERS →

deep-research.intelliforge.tech

TOPICS COVERED

- Embeddings: cosine similarity, dimensions, distance metrics
- Vector stores: pgvector, Pinecone, Chroma, Weaviate — tradeoffs
- Supabase pgvector: setup, HNSW indexing, IVFFlat strategies
- Chunking: fixed, recursive, semantic, sliding window
- Hybrid search: BM25 keyword + dense vector retrieval

SESSION DELIVERABLE

Data Pipeline

Set up Supabase with pgvector. Ingest a 50-page PDF with 3 chunking strategies, generate embeddings, run 10 semantic queries. Produce a markdown comparison table of retrieval quality across chunk sizes.

↑ PRE-READ RESOURCES

Complete these before the session

01

What Are Embeddings? — Vicki Boykis

https://vickiboykis.com/what_are_embeddings/

02

Supabase Vector & AI Guide

<https://supabase.com/docs/guides/ai/vector-columns>

03

Chunking Strategies — Pinecone Blog

<https://www.pinecone.io/learn/chunking-strategies/>

↓ POST-READ RESOURCES

Deepen your understanding after

01

pgvector GitHub — README & indexing benchmarks

<https://github.com/pgvector/pgvector>

02

Sentence Transformers — HuggingFace

<https://www.sbert.net/>

03

Chroma Getting Started Guide

<https://docs.trychroma.com/getting-started>

W04

RAG Systems Architecture

Build retrieval-augmented pipelines that work in production

PHASE 2 · RAG & DATA

POWERS →

deep-research.intelliforge.tech

TOPICS COVERED

- Naive RAG vs Advanced RAG vs Modular RAG — comparison
- Query transformation: HyDE, step-back prompting, multi-query
- Re-ranking: cross-encoders, Cohere Rerank, reciprocal rank fusion
- Context compression, stuffing, lost-in-the-middle problem
- RAG evaluation: RAGAS — faithfulness, relevancy, context recall

SESSION DELIVERABLE

Portfolio: RAG + Generator

Portfolio P3 — Research-to-Deck Generator: RAG over 50+ papers via Semantic Scholar API → synthesise findings → auto-generate branded PPTX with citations using python-pptx. Next.js API route. Deploy to Vercel. Share live URL. This is the core pipeline powering **deep-research.intelliforge.tech**.

↑ PRE-READ RESOURCES

Complete these before the session

- 01 RAG Survey — Gao et al. 2023 (arXiv)
<https://arxiv.org/abs/2312.10997>
- 02 RAGAS Evaluation Framework Docs
<https://docs.ragas.io/en/latest/>
- 03 HyDE Paper (arXiv)
<https://arxiv.org/abs/2212.10496>

↓ POST-READ RESOURCES

Deepen your understanding after

- 01 LlamaIndex RAG Tutorial
https://docs.llamaindex.ai/en/stable/getting_started/starter_example/
- 02 Cohere Rerank API Reference
<https://docs.cohere.com/reference/rerank>
- 03 Advanced RAG Techniques — Towards AI
<https://towardsai.net/p/machine-learning/advanced-rag-techniques>

W05

PHASE 3 · AGENTS & AUTOMATION

AI Agent Design Patterns

From single LLM calls to autonomous multi-step agents

POWERS →

• upskill.intelliforge.tech

TOPICS COVERED

→ ReAct pattern: Reasoning + Acting loops with tool use

→ Tool/function calling: schema definition, execution, result injection

→ Agent memory: in-context, external (Redis/Supabase), episodic

→ Multi-agent orchestration: supervisor, swarm, pipeline patterns

→ Error handling, retry logic, guardrails and safety patterns

SESSION DELIVERABLE

Portfolio: Agentic System

Portfolio P1 — Compliance Sentinel Agent: LangGraph agent scrapes regulatory sites daily → audits codebase against new rules → generates audit-ready PDF report. Playwright + Legal RAG + BullMQ cron on Fly.io. Public GitHub + Loom demo.

↑ PRE-READ RESOURCES

Complete these before the session

01

ReAct Paper — arXiv

<https://arxiv.org/abs/2210.03629>

02

Anthropic: Building Effective Agents

<https://www.anthropic.com/research/building-effective-agents>

03

LangGraph Conceptual Guide

<https://langchain-ai.github.io/langgraph/concepts/>

↓ POST-READ RESOURCES

Deepen your understanding after

01

OpenAI Swarm — Multi-Agent Framework (GitHub)

<https://github.com/openai/swarm>

02

AutoGen — Microsoft Multi-Agent Docs

<https://microsoft.github.io/autogen/>

03

CrewAI Documentation

<https://docs.crewai.com/>

W06

PHASE 3 · AGENTS & AUTOMATION

Task Queues & Async AI Workflows

Production-grade background processing for AI workloads

POWERS →

• upskill.intelliforge.tech

TOPICS COVERED

- BullMQ architecture: queues, workers, jobs, concurrency, priorities
- Long-running AI job patterns: webhooks, polling, Server-Sent Events
- Rate limiting and exponential backoff for LLM API calls
- Fly.io Mumbai worker deployment: Dockerfile, secrets, auto-scaling
- Observability: job dashboards, retry policies, dead-letter queues

SESSION DELIVERABLE

Portfolio: Multi-Agent

Portfolio P2 — Autonomous Code Reviewer: 4 specialised agents (Security, Style, Coverage, Architecture) debate every PR in parallel. GitHub Actions webhook → BullMQ → LLM-as-judge verdict → GitHub review comment. Fly.io deploy.

↑ PRE-READ RESOURCES

Complete these before the session

01

BullMQ Official Documentation
<https://docs.bullmq.io/>

02

Fly.io Getting Started
<https://fly.io/docs/getting-started/>

03

Server-Sent Events — MDN
https://developer.mozilla.org/en-US/docs/Web/API/Server-sent_events

↓ POST-READ RESOURCES

Deepen your understanding after

01

Redis University RU101 (free course)
<https://university.redis.com/courses/ru101/>

02

Inngest — Durable Functions for TypeScript
<https://www.inngest.com/docs>

03

Trigger.dev — Background Jobs Docs
<https://trigger.dev/docs>

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W07

PHASE 4 · FULL-STACK AI

Next.js 14 + AI Integration

Build AI-native web apps with App Router and streaming UIs

POWERS →

• hrms.intelliforge.tech

TOPICS COVERED

- App Router: server components, actions, route handlers
- Vercel AI SDK: useChat, useCompletion, streaming responses
- Streaming UI: RSC, Suspense boundaries, skeleton states
- Clerk: multi-tenant org-scoped auth patterns
- AI SDK tool calling + generative UI (experimental)

SESSION DELIVERABLE

Portfolio: Swarm + SaaS

Portfolio P4 — Recruiting Swarm: NER resume parser → pgvector candidate-JD scoring → Claude drafts personalised outreach → Google Calendar books slots. Next.js 14 + Clerk recruiter dashboard. Deploy to Vercel. This is the agentic hiring pipeline behind **hrms.intelliforge.tech**.

↑ PRE-READ RESOURCES

Complete these before the session

01

Next.js App Router Documentation
<https://nextjs.org/docs/app>

02

Vercel AI SDK Documentation
<https://sdk.vercel.ai/docs>

03

Clerk Next.js Quickstart
<https://clerk.com/docs/quickstarts/nextjs>

↓ POST-READ RESOURCES

Deepen your understanding after

01

Next.js AI Chatbot Template — Vercel (GitHub)
<https://github.com/vercel/ai-chatbot>

02

shadcn/ui Component Library
<https://ui.shadcn.com/>

03

React Server Components Deep Dive — Josh W Comeau
<https://www.joshwcomeau.com/react/server-components/>

W08

PHASE 4 · FULL-STACK AI

Prisma, Supabase & Payments

Data layer design and monetisation for AI-native products

POWERS →

- hrms.intelliforge.tech
- upskill.intelliforge.tech

TOPICS COVERED

- Prisma schema for AI apps: usage tracking, multi-tenancy
- Supabase Row Level Security for tenant data isolation
- Razorpay: subscription plans, webhook handlers, paise arithmetic
- Usage-based billing: token counting, quota enforcement
- DB migrations, seed scripts, CI/CD schema deploy patterns

SESSION DELIVERABLE

Monetised App

Add Free/Pro tiers to your W07 app via Razorpay. Token quota enforced through Prisma usage table. Webhook handler updates subscription state. All amounts in paise. Include a working Prisma seed script.

↑ PRE-READ RESOURCES

Complete these before the session

- 01 Prisma Getting Started — Official Docs
<https://www.prisma.io/docs/getting-started>
- 02 Supabase Row Level Security Guide
<https://supabase.com/docs/guides/auth/row-level-security>
- 03 Razorpay Subscriptions API
<https://razorpay.com/docs/payments/subscriptions/>

↓ POST-READ RESOURCES

Deepen your understanding after

- 01 Prisma with Supabase — Official Guide
<https://www.prisma.io/docs/guides/database/supabase>
- 02 Stripe Billing Patterns (applicable to Razorpay)
<https://stripe.com/docs/billing/subscriptions/overview>
- 03 PlanetScale Schema Branching (migration inspiration)
<https://planetscale.com/docs/concepts/branching>

W09

PHASE 5 · PRODUCTION & DEVOPS

Production Deployment & Observability

Ship AI apps that don't break at 2 AM

POWERS →

- upskill.intelliforge.tech
- hrms.intelliforge.tech

TOPICS COVERED

- Vercel: env vars, preview deploys, edge config, regions
- Fly.io: multi-region Dockerfiles, volumes, secrets, scaling
- OpenTelemetry instrumentation for Next.js + AI workloads
- Structured logging: request IDs, org context, LLM latency tags
- Health checks, circuit breakers, graceful degradation

SESSION DELIVERABLE

Runbook + Monitoring

Add full observability to your deployed app: Sentry error tracking, Axiom structured logs with LLM latency metrics, /healthz endpoint. Write a runbook (markdown) covering the 3 most likely failure modes and remediation steps.

↑ PRE-READ RESOURCES

Complete these before the session

01

Vercel Deployment Overview
<https://vercel.com/docs/deployments/overview>

02

Fly.io App Deploy Guide
<https://fly.io/docs/apps/deploy/>

03

OpenTelemetry for Node.js
<https://opentelemetry.io/docs/languages/js/getting-started/nodejs/>

↓ POST-READ RESOURCES

Deepen your understanding after

01

Axiom — Log Management (free tier)
<https://axiom.co/docs>

02

Sentry Next.js SDK
<https://docs.sentry.io/platforms/javascript/guides/nextjs/>

03

The Twelve-Factor App
<https://12factor.net/>

W10

PHASE 5 · PRODUCTION & DEVOPS

Security, Cost & Responsible AI

AI-specific security patterns and governance for production

POWERS →

- deep-research.intelliforge.tech
- hrms.intelliforge.tech

TOPICS COVERED

- Prompt injection attacks: anatomy, detection, mitigation
- PII handling: redaction and masking middleware before LLM calls
- API key lifecycle: rotation, scoping, secrets managers
- LLM cost governance: semantic caching, model routing, fallbacks
- Responsible AI: output filtering, content moderation, audit logs

↑ PRE-READ RESOURCES

Complete these before the session

01

OWASP LLM Top 10
<https://owasp.org/www-project-top-10-for-large-language-model-applications/>

02

Prompt Injection Explained — Simon Willison
<https://simonwillison.net/2022/Sep/12/prompt-injection/>

03

Anthropic Responsible Scaling Policy
<https://www.anthropic.com/index/anthropics-responsible-scaling-policy>

SESSION DELIVERABLE

Portfolio: LLM Gateway

Portfolio P5 — Cost-Aware Agent Router: LLM gateway with complexity classifier → dynamic model routing (Haiku→Sonnet→Opus) → per-org token budget enforcement → Prometheus metrics dashboard. 60% cost reduction. Drop-in middleware.

↓ POST-READ RESOURCES

Deepen your understanding after

01

LLM Guard — Open Source Security Toolkit
<https://llm-guard.com/>

02

Portkey — LLM Gateway & Cost Control
<https://portkey.ai/docs>

03

Azure AI Content Safety (reference)
<https://learn.microsoft.com/en-us/azure/ai-services/content-safety/>

W11

PHASE 6 · CAPSTONE

Capstone — Build & Ship

End-to-end AI-native SaaS product from brief to deployment

POWERS →

- upskill.intelliforge.tech
- learning.intelliforge.tech

TOPICS COVERED

→ Product brief selection: 5 IntelliForge-curated briefs

→ Architecture design: system diagram, data model, AI pipeline flow

→ Full build sprint: frontend, API routes, AI pipeline, background jobs

→ Peer code review: structured session with 2 cohort partners

→ Demo prep: 3-min Loom + GitHub README with architecture notes

SESSION DELIVERABLE

Shipped Product

A fully deployed AI SaaS product: live URL, public GitHub repo, 3-min Loom demo, architecture diagram, README covering stack choices and AI design decisions. Presented live to cohort for peer review. Your capstone is featured on your upskill.intelliforge.tech profile and learning.intelliforge.tech cohort showcase.

↑ PRE-READ RESOURCES

Complete these before the session

01

YC Startup School — Building Product (free)
<https://www.startupschool.org/>

02

System Design Primer (GitHub)
<https://github.com/donnemartin/system-design-primer>

03

Architecture Decision Records — Michael Nygard
<https://cognitect.com/blog/2011/11/15/documenting-architecture-decisions>

↓ POST-READ RESOURCES

Deepen your understanding after

01

Make a README — Best Practices
<https://www.makeareadme.com/>

02

Product Hunt Launch Checklist
<https://www.producthunt.com/posts/ph-launch-checklist>

03

Show HN — Hacker News Launch Guide
<https://news.ycombinator.com/showhn.html>

W12

PHASE 6 · CAREER & GTM

Career Strategy & Personal GTM

Position as an AI engineer, not just someone who uses AI tools

POWERS →

- upskill.intelliforge.tech
- learning.intelliforge.tech

TOPICS COVERED

- LinkedIn profile for AI roles: headline, about, featured section
- Portfolio: GitHub profile README, personal site, case studies
- Freelance path: Topmate, direct acquisition, rate setting
- FTE path: AI-first companies in India, negotiation, equity basics
- Open source: IntelliForge OSS projects + community network

SESSION DELIVERABLE

Personal Brand

Ship your personal brand: updated LinkedIn (AI engineer positioning), GitHub profile README showcasing 11-week portfolio, Topmate or personal site page. Share links in cohort channel for peer feedback.

↑ PRE-READ RESOURCES

Complete these before the session

01

Developer Portfolio Guide — freeCodeCamp
<https://www.freecodecamp.org/news/how-to-build-a-developer-portfolio-website/>

02

Salary Negotiation — levels.fyi India
<https://www.levels.fyi/t/software-engineer/location/India>

03

How to Contribute to Open Source — GitHub
<https://opensource.guide/how-to-contribute/>

↓ POST-READ RESOURCES

Deepen your understanding after

01

Topmate Creator Guide
<https://topmate.io/blog/creator-guide>

02

Wellfound (AngelList India) — AI Jobs
<https://wellfound.com/jobs>

03

IntelliForge GitHub Org
<https://github.com/intelliforge-ai>

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P7

Phase 7 · Portfolio Projects

Most bootcamp graduates build chatbots. These 5 projects are engineered differently — each solves an expensive real-world problem, carries a quantified business impact metric, and demonstrates the systems thinking that separates an **Agentic AI Engineer** from a prompt engineer.

THE PRINCIPLE

"Most people build chatbots. **Builders build systems that solve expensive problems.**"

Every project targets a high-cost business problem with measurable ROI — the difference between a portfolio that gets interviews and one that doesn't.

P1 Compliance Sentinel Agent

Saves legal teams 40+ hours/week

Autonomous regulatory monitoring & code audit system

Stack: LangGraph · Playwright · pgvector + RAG · BullMQ + Fly.io · WeasyPrint

P2 Autonomous Code Reviewer

Cuts code review time by 70%

4-agent PR debate system with LLM-as-judge verdict

Stack: GitHub Actions · Tree-sitter · LLM-as-judge · BullMQ · GitHub REST API

P3 Research-to-Deck Generator

Turns 3 days of research into 5 minutes

RAG pipeline that turns 50+ papers into an editable PowerPoint

Stack: Semantic Scholar API · pgvector + RAG · python-pptx · BullMQ · Next.js + Vercel

P4 Recruiting Swarm

Increases interview show-up rates by 40%

End-to-end AI hiring pipeline from resume to booked interview

Stack: spaCy / Gliner · pgvector · SendGrid · Google Calendar API · Next.js + Clerk

P5 Cost-Aware Agent Router

Reduces inference costs by 60%

Intelligent LLM gateway with token budgeting & dynamic model routing

Stack: Hono / Express · Claude API · Prisma + Postgres · Prometheus · Grafana

P1

Compliance Sentinel Agent

P1

PORTFOLIO PROJECT

Compliance Sentinel Agent

Autonomous regulatory monitoring & code audit system

BUSINESS IMPACT

Saves legal teams 40+ hours/week

Builds on W05 (Agent Design) + W03 (RAG) + W06 (BullMQ)

upskill.intelliforge.tech

WHAT IT DOES

→ Scrapes regulatory portals daily for new rules and amendments

→ Parses and embeds regulations into a Legal RAG vector store

→ LangGraph agent audits target codebase against updated rules

→ Generates structured, audit-ready PDF reports with citations

→ Slack/email alerts when critical compliance violations detected

TECH STACK

Every tool chosen for production relevance

LangGraph

Agent orchestration & multi-step reasoning

Playwright

Headless browser for regulatory site scraping

pgvector + RAG

Legal document retrieval & semantic search

BullMQ + Fly.io

Daily scheduled cron job, Mumbai region

WeasyPrint

Audit-ready PDF report generation

ARCHITECTURE BREAKDOWN

Crawler

Playwright headless browser scrapes regulatory sites on BullMQ cron schedule

RAG Layer

Regulations chunked + embedded into Supabase pgvector for semantic retrieval

Agent

LangGraph: plan → retrieve rules → analyse code → flag violations → draft report

Report

WeasyPrint generates branded audit PDF with violation table and citations

Notify

Slack webhook + SendGrid email on critical violation detection

HIRING SIGNALS DEMONSTRATED

✓ Agentic reasoning with multi-step tool use

✓ Production RAG over domain-specific legal corpus

✓ Headless browser automation in an AI pipeline

✓ Scheduled workers with failure recovery

✓ Business impact quantification

P2

Autonomous Code Reviewer

P2

PORTFOLIO PROJECT

Autonomous Code Reviewer

4-agent PR debate system with LLM-as-judge verdict

BUSINESS IMPACT

Cuts code review time by 70%

Builds on W06 (Async Queues) + W05 (Agents) + W02 (LLM-as-judge)

[upskill.intelliforge.tech](#)

WHAT IT DOES

→ Triggered automatically on every GitHub Pull Request via webhook

→ Spins up 4 specialised agents in parallel per PR diff

→ Each agent analyses from its domain: Security, Style, Tests, Architecture

→ LLM-as-judge aggregates verdicts into a final consensus review

→ Posts structured review comment directly to the GitHub PR thread

TECH STACK

Every tool chosen for production relevance

GitHub Actions

Webhook trigger & CI integration

Tree-sitter

AST parsing for structured diff analysis

LLM-as-judge

Claude-based verdict aggregation

BullMQ

Parallel agent queue with concurrency control

GitHub REST API

Post review comments programmatically

ARCHITECTURE BREAKDOWN

Trigger

GitHub Actions webhook fires on pull_request, queues job in BullMQ

Parser

Tree-sitter parses diff into AST for structured code analysis

Agent Swarm

4 parallel agents: Security · Style & Lint · Test Coverage · Architecture

Judge

Claude LLM-as-judge reads 4 verdicts, synthesises consensus, assigns severity

Output

GitHub REST API posts structured review comment with per-agent breakdown

HIRING SIGNALS DEMONSTRATED

✓ Multi-agent parallel orchestration pattern

✓ AST-based code understanding beyond raw text

✓ LLM-as-judge evaluation methodology

✓ GitHub Actions & webhook integration

✓ Structured output enforcement across agent ensemble

P3

Research-to-Deck Generator

BUSINESS IMPACT

P3

PORTFOLIO PROJECT

Research-to-Deck Generator

RAG pipeline that turns 50+ papers into an editable PowerPoint

Turns 3 days of research into 5 minutes

Builds on W04 (RAG Architecture) + W03 (Embeddings) + W06 (Async)

deep-research.intelliforge.tech

WHAT IT DOES

- Accepts a topic and fetches 50+ papers via Semantic Scholar API
- Chunks and embeds all papers into a topic-scoped vector store
- Multi-query RAG synthesises key findings, trends, and gaps
- Generates structured slide outline with per-claim citations
- Exports a fully editable PPTX with speaker notes and branding

ARCHITECTURE BREAKDOWN

Ingestion	Semantic Scholar API → fetch PDFs → chunk → embed → pgvector
RAG	Multi-query RAG with re-ranking to surface highest-signal findings
Synthesis	Claude generates slide titles, bullets, citation mappings
PPTX	python-pptx assembles branded deck: title, agenda, content, citations
API	Next.js route handler: POST topic → BullMQ job → download link

TECH STACK

Every tool chosen for production relevance

Semantic Scholar API	Free academic paper search & fetch
pgvector + RAG	Multi-query retrieval over paper corpus
python-pptx	Programmatic PPTX generation with branding
BullMQ	Async generation job — papers take 2–3 min
Next.js + Vercel	Frontend with job polling and download

HIRING SIGNALS DEMONSTRATED

- ✓ Academic API integration at scale (50+ papers)
- ✓ Multi-document RAG with citation tracking
- ✓ Programmatic document generation
- ✓ Background job pattern for long-running AI tasks
- ✓ Full-stack AI product from API to file download

P4

Recruiting Swarm

P4

PORTFOLIO PROJECT

Recruiting Swarm

End-to-end AI hiring pipeline from resume to booked interview

BUSINESS IMPACT

Increases interview show-up rates by 40%

Builds on W07 (Next.js + Full-Stack) + W03 (Embeddings) + W05 (Agents)

hrms.intelliforge.tech

WHAT IT DOES

→ NER model parses uploaded resumes into structured candidate profiles

→ Scores candidate fit against a JD via vector similarity

→ Claude drafts personalised outreach emails per candidate

→ Google Calendar API auto-books interview slots

→ Recruiter dashboard: kanban pipeline, approve/reject, email preview

TECH STACK

Every tool chosen for production relevance

spaCy / Gliner

NER-based resume information extraction

pgvector

Candidate-to-JD semantic similarity scoring

SendGrid

Transactional outreach email delivery

Google Calendar API

Automated interview slot booking

Next.js + Clerk

Multi-tenant recruiter dashboard

ARCHITECTURE BREAKDOWN

Parser

spaCy/Gliner NER extracts name, skills, exp, education from PDF

Scoring

Candidate embedding vs JD embedding — cosine similarity fit score

Outreach

Claude drafts personalised email using extracted candidate profile

Calendar

Google Calendar API finds open slots, sends invite, updates record

Dashboard

Next.js recruiter UI: kanban pipeline, approve/reject, email preview

HIRING SIGNALS DEMONSTRATED

✓ NER-based document understanding

✓ Vector similarity for matching and ranking

✓ Multi-step agent with external API integrations

✓ Email automation with personalisation at scale

✓ Full-stack multi-tenant SaaS with approval workflow

P5

Cost-Aware Agent Router

P5

PORTFOLIO PROJECT

Cost-Aware Agent Router

Intelligent LLM gateway with token budgeting & dynamic model routing

BUSINESS IMPACT

Reduces inference costs by 60%

Builds on W10 (Security & Cost) + W09 (Observability) + W06 (Async)

[deep-research.intelliforge.tech](#)

WHAT IT DOES

→ Intercepts all LLM API calls as a drop-in proxy middleware

→ Classifies query complexity to route to cheapest capable model

→ Enforces per-request and per-org token budgets via Prisma

→ Exits early when confidence threshold met — skips expensive models

→ Exposes Prometheus metrics: cost/request, latency, model distribution

TECH STACK

Every tool chosen for production relevance

Hono / Express

Lightweight proxy middleware layer

Claude API

Haiku → Sonnet → Opus tiered routing

Prisma + Postgres

Token budget tracking per org

Prometheus

Real-time cost & latency metrics

Grafana

Dashboard: savings, model distribution, budget burn

ARCHITECTURE BREAKDOWN

Proxy

Hono/Express middleware intercepts requests before LLM APIs

Classifier

Fast Haiku call classifies complexity: simple/moderate/complex

Router

Routes: simple→Haiku, moderate→Sonnet, complex→Opus

Budget

Prisma tracks token usage per org/user; rejects over-budget requests

Metrics

Prometheus: tokens_used, cost_usd, model_selected, latency_ms

HIRING SIGNALS DEMONSTRATED

✓ LLM gateway / proxy architecture pattern

✓ Dynamic model selection by complexity

✓ Budget enforcement at the infrastructure layer

✓ Prometheus metrics for AI workloads

✓ Cost optimisation as first-class engineering

P8

Phase 8 · Open-Source Takeaways

The fastest way to get hired as an AI engineer is to have your name in a production codebase that hiring managers already use. These 12 open-source projects are chosen because they map directly to curriculum weeks, align with IntelliForge product stack, and have active maintainer communities that welcome first-time contributors. Each entry shows exactly what to contribute and why it matters.

CONTRIBUTION STRATEGY

Don't contribute randomly. Pick **one project per phase** that matches your week deliverable. A merged PR on LangGraph during W05 week carries more weight than 10 closed issues. Two IntelliForge OSS projects (PDFForge, @intelliforge/otp) offer direct pipeline into the IntelliForge product team.

AI ROADMAP COHORT — WHATSAPP GROUP

Share your OSS contributions, get PR reviews, and find pair-programming partners. Scan QR or visit the link to join.
chat.whatsapp.com/LwxMCJIEqLm4oLOG0fLqmE

PHASE 1

LiteLLM

Python

13k+

W02

upskill

Call 100+ LLMs using the OpenAI format

github.com/BerriAI/litellm

Instructor

Python

9k+

W02

upskill

Structured outputs from LLMs using Pydantic

github.com/jxnl/instructor

PHASE 2

RAGAS

Python

7k+

W04

deep-research

Evaluation framework for RAG pipelines

github.com/explodinggradients/ragas

Chroma

Python / Rust

15k+

W03

deep-research

Open-source AI-native vector database

github.com/chroma-core/chroma

PHASE 3

LangGraph

Python / TypeScript

8k+

W05

upskill

Build stateful multi-actor agent applications

github.com/langchain-ai/langgraph

CrewAI Python 26k+
Framework for orchestrating role-playing AI agents
github.com/crewAIInc/crewAI

W05
hrms

PHASE 4

Vercel AI SDK TypeScript 12k+
TypeScript toolkit for building AI-powered apps
github.com/vercel/ai

W07
upskill

Trigger.dev TypeScript 9k+
Open-source background jobs for TypeScript
github.com/triggerdotdev/trigger.dev

W06
upskill

PHASE 5

LLM Guard Python 3k+
Open-source security toolkit for LLM interactions
github.com/protectai/llm-guard

W10
hrms

Portkey Gateway TypeScript 6k+
Open-source LLM gateway with fallbacks & observability
github.com/Portkey-AI/gateway

W10
deep-research

PHASE 6

PDFForge Python / TypeScript IntelliForge OSS
Self-hosted AI PDF toolkit with MCP server
github.com/intelliforge-ai/pdfforge

W11
learning

@intelliforge/otp TypeScript IntelliForge OSS
Production-grade phone OTP with WhatsApp/SMS fallback
github.com/intelliforge-ai/otp

W07
upskill

→

Contribution Deep Dive

YOUR CONTRIBUTION ROADMAP

Week 1–4: Star + clone all repos · Read CONTRIBUTING.md · Pick one "good first issue" · **Week 5–8:** Open draft PR on chosen project · Get review + merge · **Week 9–12:** Second PR on IntelliForge OSS · Add to portfolio README · **Post-W12:** Maintain your contribution, mentor others, build your contributor profile

LiteLLM

Python

13k+

W02

upskill

Call 100+ LLMs using the OpenAI format

Drop-in multi-LLM router — perfect for understanding SDK abstractions. Study the routing logic for P5 Cost-Aware Agent Router.

HOW TO CONTRIBUTE

- Add a new LLM provider adapter
- Write cost-per-token tests
- Improve TypeScript type definitions

<https://github.com/BerriAI/litellm>

Vercel AI SDK

TypeScript

12k+

W07

upskill

TypeScript toolkit for building AI-powered apps

Used directly in W07. Contributing to this repo puts your name in a library that millions of Next.js developers will use.

HOW TO CONTRIBUTE

- Add a Razorpay payment example
- Write Supabase streaming integration docs
- Fix open TypeScript type issues

<https://github.com/vercel/ai>

Instructor

Python

9k+

W02

upskill

Structured outputs from LLMs using Pydantic

The canonical library for validated LLM outputs. Maps directly to W02 structured output patterns and Zod validation in TypeScript.

HOW TO CONTRIBUTE

- Add an example for a new domain (legal, medical)
- Write a tutorial notebook
- Fix open validator issues

<https://github.com/jxnl/instructor>

Trigger.dev

TypeScript

9k+

W06

upskill

Open-source background jobs for TypeScript

Alternative to BullMQ studied in W06. Contributing demonstrates TypeScript background job architecture — key hiring signal.

HOW TO CONTRIBUTE

- Build an AI job example (LLM batch processing)
- Add Fly.io deployment guide
- Write Prisma integration tutorial

<https://github.com/triggerdotdev/trigger.dev>

RAGAS

Python

7k+

W04

deep-research

Evaluation framework for RAG pipelines

Used directly in W04 RAG eval deliverable. Contributing here gives you a published open-source eval credential.

HOW TO CONTRIBUTE

- Add a new eval metric (e.g. toxicity)
- Improve async evaluation support
- Write integration docs for Supabase pgvector

<https://github.com/explodinggradients/ragas>

LLM Guard

Python

3k+

W10

hrms

Open-source security toolkit for LLM interactions

Core tool in W10 security audit deliverable. Adding a new scanner here is a publishable contribution to AI safety.

HOW TO CONTRIBUTE

- Add a new PII scanner (Aadhaar, Indian phone formats)
- Write a Next.js middleware integration
- Benchmark scanner performance

<https://github.com/protectai/llm-guard>

Chroma

Python / Rust

W03

deep-research

15k+

Open-source AI-native vector database

Foundational vector store studied in W03. Contributing here demonstrates deep understanding of embedding infrastructure.

HOW TO CONTRIBUTE

- Add a distance metric variant
- Write a Next.js integration example
- Improve Docker Compose setup for self-hosting

<https://github.com/chroma-core/chroma>

Portkey Gateway

TypeScript

6k+

W10

deep-research

Open-source LLM gateway with fallbacks & observability

Directly maps to P5 Cost-Aware Agent Router. Study the routing logic, then build your own or contribute to theirs.

HOW TO CONTRIBUTE

- Add Razorpay budget enforcement middleware
- Write Prometheus metrics integration
- Add Claude Haiku/Sonnet/Opus routing config

<https://github.com/Portkey-AI/gateway>

LangGraph

Python / TypeScript

W05

upskill

8k+

Build stateful multi-actor agent applications

Core orchestration layer for P1 Compliance Sentinel Agent. Contributing here is the strongest signal for agentic AI roles.

HOW TO CONTRIBUTE

- Build a new example agent (compliance, recruiting)
- Add TypeScript equivalents for Python examples
- Improve streaming support documentation

<https://github.com/langchain-ai/langgraph>

PDFForge

Python / TypeScript

W11

learning

IntelliForge OSS

Self-hosted AI PDF toolkit with MCP server

IntelliForge's own open-source project. Contributing here gets you directly into the IntelliForge talent network and product team pipeline.

HOW TO CONTRIBUTE

- Add a new PDF extraction tool
- Build an MCP server integration
- Write Next.js SDK examples

<https://github.com/intelliforge-ai/pdfforge>

CrewAI

Python

26k+

W05

hrms

Framework for orchestrating role-playing AI agents

Role-based agent framework studied in W05. Maps to P4 Recruiting Swarm architecture. High stars = strong portfolio signal.

HOW TO CONTRIBUTE

- Add an HR recruiting crew example
- Write a BullMQ task queue integration
- Improve tool error handling docs

<https://github.com/crewAIInc/crewAI>

@intelliforge/otp

TypeScript

W07

upskill

IntelliForge OSS

Production-grade phone OTP with WhatsApp/SMS fallback

Used across IntelliForge products. Adding your name to this package means your code runs in production on live IntelliForge platforms.

HOW TO CONTRIBUTE

- Add an email OTP fallback channel
- Write Prisma usage tracking integration
- Add rate limiting middleware

<https://github.com/intelliforge-ai/otp>